**Difference Between Digital Certificate and Digital Signature**

| Feature | Digital Certificate | Digital Signature |
|---|---|---|
| Definition | A document used to verify the identity of a user or website | A technique used to verify authenticity and integrity of a message |
| Purpose | Provides identity authentication | Ensures message is not altered |
| Issued by | Certificate Authority (CA) | Created by sender using private key |
| Contains | Public key, owner details, CA signature | Encrypted hash value of message |
| Security Goal | Authentication | Authentication + Integrity + Non-repudiation |
| Usage | Used in HTTPS, SSL/TLS | Used in emails, documents, transactions |

## ✅ SHA-1 vs MD5 (Table – 8 Points)

| No. | Feature | MD5 | SHA-1 |
|---|---|---|---|
| 1 | Full Form | Message Digest 5 | Secure Hash Algorithm 1 |
| 2 | Output Size | 128-bit | 160-bit |
| 3 | Security Level | Very weak | Stronger than MD5 (but now weak) |
| 4 | Collision Resistance | Easily broken | Better but vulnerable |
| 5 | Speed | Faster | Slower |
| 6 | Structure | Simpler algorithm | More complex algorithm |
| 7 | Usage | Checksums, file verification | Digital signatures, certificates |
| 8 | Current Status | Deprecated | Also deprecated |

**Explain SQL injection Attack.**

### Definition

**SQL Injection is a type of cyber attack in which an attacker inserts malicious SQL queries into input fields (like login forms) to access, modify, or delete database data.**

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### How it Works

1. **A website takes user input (username/password)**
2. **Input is directly used in SQL query without validation**
3. **Attacker injects malicious SQL code**
4. **Database executes it → unauthorized access**

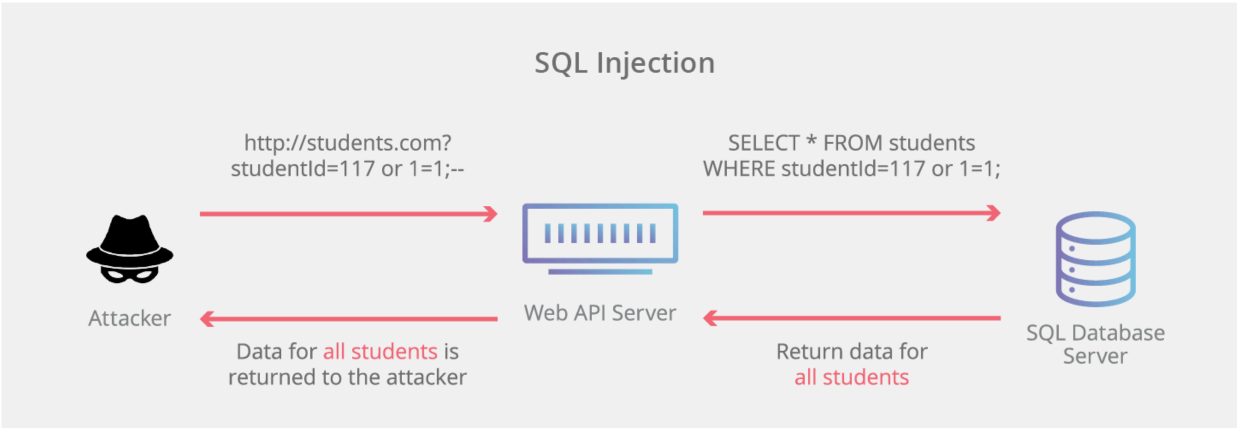
**EXAMPLE**

# Query 1

**SELECT * FROM users WHERE username='' OR 1=1;**

**OR 1=1 → Always true**

**SQL Injection uses OR  1=1**




**List Types of computer viruses.**

# ✅ Types of Computer Viruses

1. **Boot Sector Virus**
    - Attacks boot sector of hard disk
    - Activates when system starts
2. **File Virus (Program Virus)**
    - Infects executable files (.exe, .com)
3. **Macro Virus**
    - Infects documents (Word, Excel) using macros
4. **Multipartite Virus**
    - Infects both boot sector and files
5. **Polymorphic Virus**
    - Changes its code to avoid detection
6. **Resident Virus**
    - Stays in memory and infects files automatically
7. **Direct Action Virus**
    - Activates only when infected file is executed
8. **Overwrite Virus**
    - Overwrites file content, destroying data

**Explain ARP Spoofing**

**ARP Spoofing**

**Definition:**
ARP spoofing is an attack where fake ARP messages are sent to link attacker’s MAC address with a legitimate IP address.

**Working:**

1. Attacker sends fake ARP reply
2. Victim updates ARP table with wrong MAC
3. Data is redirected to attacker
4. Attacker can intercept or modify data

**Effects:**

- Man-in-the-middle attack
- Data theft
- Network disruption

**Prevention:**

- Use static ARP entries
- Use secure protocols (HTTPS, SSH)
- Use intrusion detection systems

**Conclusion:**
ARP spoofing redirects traffic using fake ARP messages, allowing data theft or modification

**Memory Trick:**
Fake ARP → Wrong MAC → Data Theft

Here is your **clean, simple notes version** 👇

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**Worms and Viruses**

**Define Worms and Viruses**

**Virus (Definition):**
A computer virus is a malicious program that attaches to a host file or program and spreads when the infected file is executed. It requires user action.

**Worm (Definition):**
A worm is a malicious program that spreads automatically across networks without a host file or user action. It can replicate itself.

**Key Difference:**

- Virus: Needs a host file and user action
- Worm: Self-spreading, no host file required

**List types of Malicious program.**

**Types of Malicious Programs (Malware)**

- Virus: Attaches to files and spreads when executed
- Worm: Spreads automatically through networks
- Trojan Horse: Looks legitimate but is harmful
- Spyware: Collects user information secretly
- Adware: Displays unwanted advertisements
- Ransomware: Locks data and demands money
- Rootkit: Hides malicious activities in system

## ✅ Difference Between HMAC and CMAC

| Feature | HMAC (Hash-based Message Authentication Code) | CMAC (Cipher-based Message Authentication Code) |
|---|---|---|
| **Full Form** | Hash-based MAC | Cipher-based MAC |
| **Based On** | Hash functions (e.g., SHA-256) | Block cipher algorithms (e.g., AES) |
| **Algorithm Type** | Uses hashing | Uses encryption |
| **Key Usage** | Secret key with hash function | Secret key with block cipher |
| **Security** | Very secure | Very secure |
| **Performance** | Faster (no encryption) | Slightly slower |
| **Implementation** | Simple | More complex |
| **Usage** | Data integrity, APIs, messages | Secure communications, encryption systems |